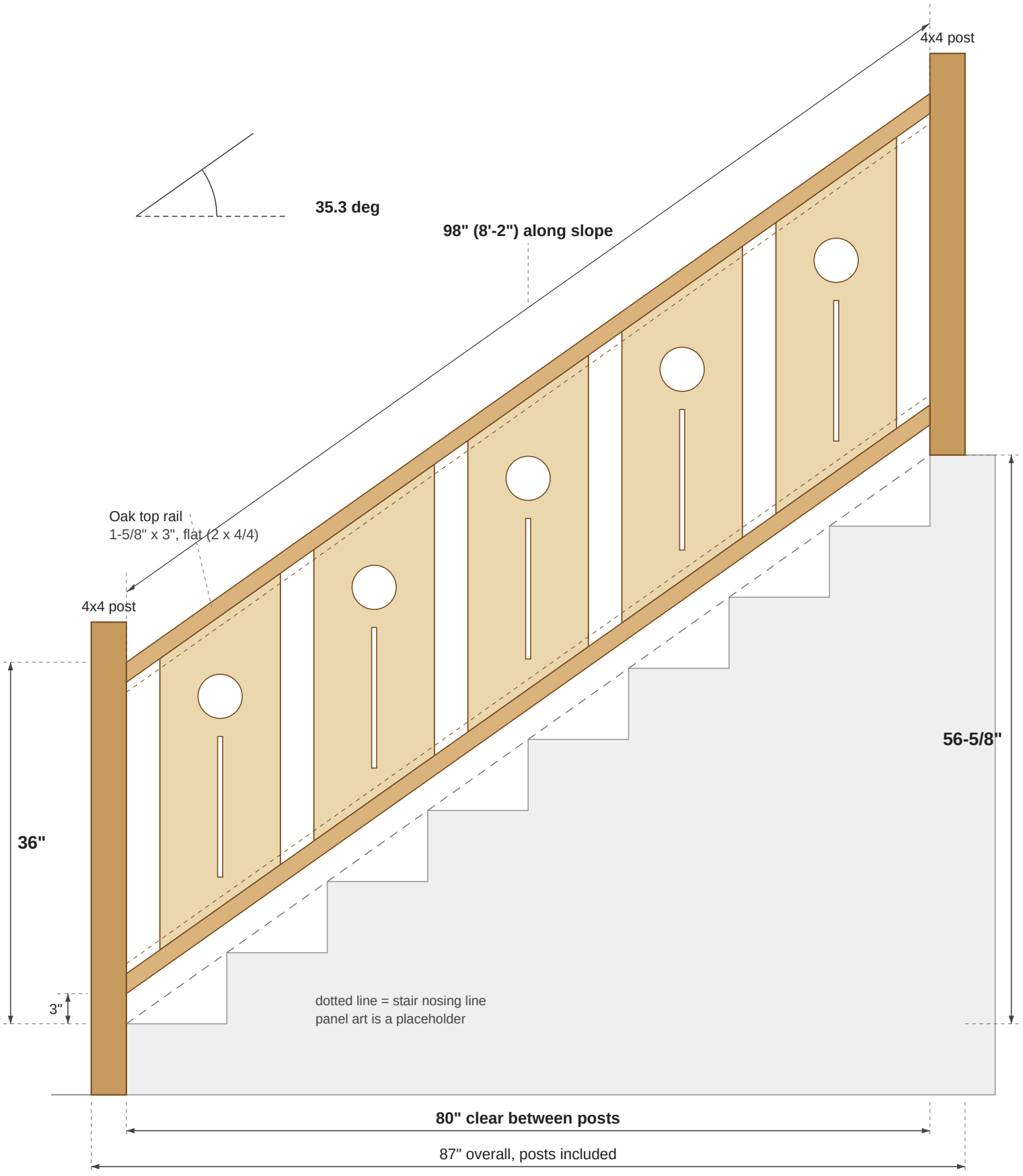
STAIR RAILING - SIDE ELEVATION

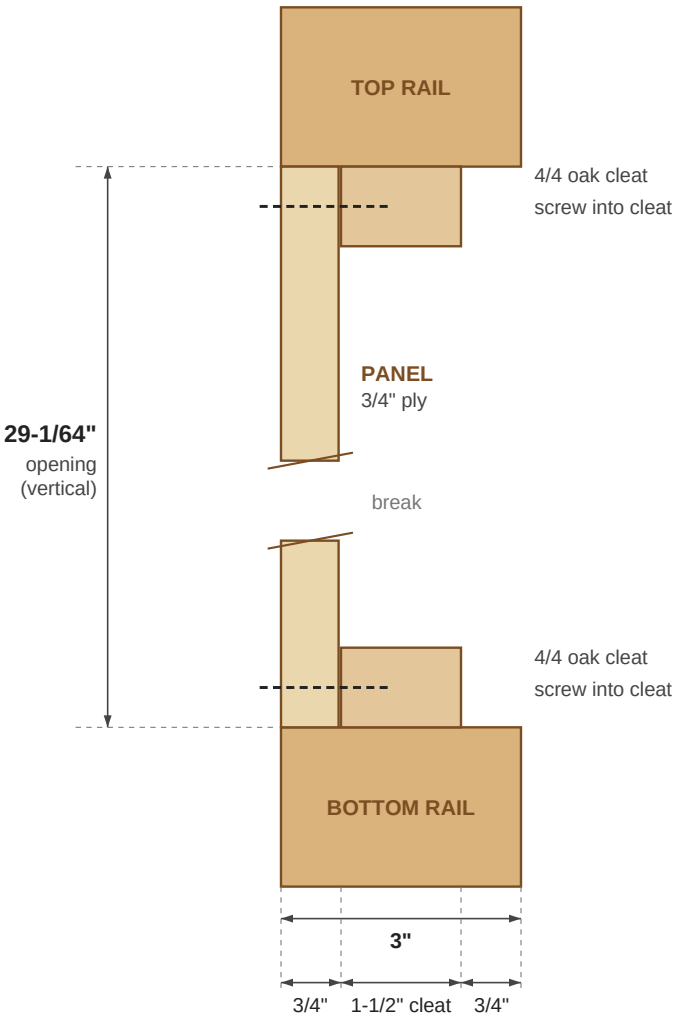
White oak S3S 4/4 (posts 8/4), cleat-mounted panel infill. Slope 35.3 deg (measured). Page 1 of 2.



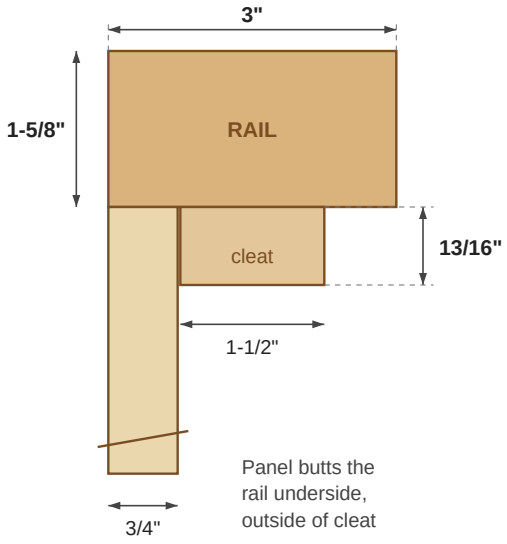
Assumes 7-1/16" riser and 10" tread (35.3 deg slope). Verify on your stair. Dashed lines show the hidden cleats behind the panels.
5 panels, 12" wide, gaps 3-1/3" each (all under 4"). Measure your real art openings against a 4" sphere.
Rails are two 4/4 boards face-glued (1-5/8"); cleats are single 4/4. Top of rail is 36" above nosing line. Use stainless fasteners (tannin).

CLEAT-MOUNTED PANEL DETAIL

Vertical section across the stair (left) and true rail section (right). Outside of stair is at left.



Top rail assembly, true section (13/16" stock)



Bottom rail is the mirror image: cleat sits on top of the rail.

Cleat placement

Set each cleat 3/4" in from the outside edge of its rail. That leaves a 3/4" ledge on the outside for the panel, so the panel face ends up flush with the rail edge.

Fasteners

Glue each cleat to its rail and add a 2" stainless screw every 12", driven up through the cleat into the rail. Fasten panels with three 1-5/8" screws into each cleat, countersunk and plugged or left as accents.

Fit

Panels go in last from the outside. Cut them 1/32" short (28-31/32") and trim to fit.

Break shortens the panel for drawing; heights are not to scale between the break lines.
Rail and cleat thicknesses read longer here (1-63/64", 63/64") because the rails slope; true sizes are on the right.

CUT LIST

Part	Qty	Size (inches)	Notes
Post, bottom	1	3-1/2 x 3-1/2 x 47-1/16	Two 8/4 glued, rip to square. Stands on floor; top 40" above nosing line.
Post, top	1	3-1/2 x 3-1/2 x 40	Stands on landing; top 40" above landing.
Rail, top and bottom	2	1-5/8 x 3 x 98	Two 4/4 glued face to face. Ends 35.3 deg off square, parallel. Laid flat.
Cleat	2	13/16 x 1-1/2 x 98	Ripped from 4/4. Ends match rails. 3/4" in from outside edge.
Panel	5	3/4 ply x 12 x 28-31/32	Vertical sides 28-31/32"; top/bottom edges 35.3 deg off square.

ASSEMBLY ORDER

1. Rails: buy 10' 4/4 (98" does not fit in 8'). Face-glue pairs, joint, then plane to 1-5/8 x 3. Rip cleats 1-1/2 wide from 4/4. Cut all ends at 35.3 deg, parallel.
2. Mark a line 3/4" in from the outside edge of each rail. Glue and screw the cleats on: under the top rail, on top of the bottom rail.
3. Anchor the posts plumb. Fit the rail-and-cleat assemblies between them. Glue each end and drive two 4-1/2" stainless screws through the post into the rail end, pre-drilled and plugged.
4. From the outside, set the panels against the cleats, trim to fit, and screw them into the cleats.
5. Add a separate graspable handrail (1-1/4" to 2" wide) at 34-38" above the nosing line if code requires one.